

Fast-reactor unit cells

When is volume mixing a good approximation to a heterogeneous lattice?

16

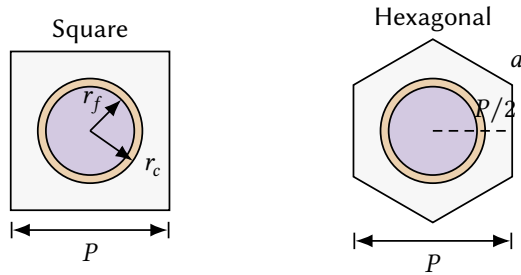
LESSON

Read: § 4.3, Eqs. (4.1)–(4.16); revisit the fast-reactor discussion in § 4.2.

1 Hexagonal Unit Cells

P is nearest-neighbor pin pitch, $D = 2r_f$ is *fuel* diameter, r_c is outer cladding radius, and L is a common axial length. A triangular array of pin centers has a hexagonal cell:

$$a = \frac{P}{\sqrt{3}}, \quad \text{apothem} = \frac{P}{2}, \quad A_{\text{hex}} = \frac{\sqrt{3}}{2} P^2, \quad A_{\text{sq}} = P^2.$$



Same pitch and pin size, different cell areas. The hexagon's side a is not its across-flats dimension P .

Omitting the gap and assigning the cladding to st ,

$$\begin{aligned} V_f &= \pi r_f^2 L, & V_{st} &= \pi(r_c^2 - r_f^2)L, \\ V_c &= (A_{\text{cell}} - \pi r_c^2)L, & V &= A_{\text{cell}}L. \end{aligned}$$

Thus, for the hexagonal cell,

$$\frac{V_f}{V} = \frac{\pi}{2\sqrt{3}} \left(\frac{D}{P} \right)^2, \quad P > 2r_c$$

for a nonzero inter-pin coolant gap. Extra structure or a fuel-clad gap must be counted separately.

2 Compare length scales, not just energies

Mean free path and optical thickness (for characteristic dimension d):

$$\lambda(E) = \frac{1}{\Sigma_t(E)}, \quad \tau(E) = d\Sigma_t(E).$$

Fast-energy cross sections are often small. For $d \ll \lambda$, spatial flux differences are modest:

$$\varphi_f(E) \approx \varphi_c(E) \approx \varphi_{st}(E).$$

A fast spectrum is generally complex and shaped by inelastic scattering processes.

3 Collapse only after finding the spectrum

For a common spectrum $\varphi(E)$, define

$$\phi = \int_0^\infty \varphi(E) dE, \quad \bar{\Sigma}_x = \frac{\int_0^\infty \Sigma_x(E) \varphi(E) dE}{\phi}.$$

Volume mixing then gives $\Sigma_x = \sum_y (V_y/V) \Sigma_{xy}$. The chapter's production/absorption balance becomes

$$k_\infty \approx \frac{V_f \bar{\Sigma}_f^f}{V_f \bar{\Sigma}_a^f + V_c \bar{\Sigma}_a^c + V_{st} \bar{\Sigma}_a^{st}}.$$

Eqs. (4.2)–(4.8); leakage and nonfission neutron multiplication neglected.

4 Enrichment is the fast-lattice lever

In the two-nuclide model (fi : fissile; fe : fertile),

$$N_f = N_{fi} + N_{fe}, \quad \tilde{e} = \frac{N_{fi}}{N_f}.$$

Let $a_i = \bar{\sigma}_a^i$, $b_i = \bar{\nu} \bar{\sigma}_f^i$ and

$$C = \frac{V_c \bar{\Sigma}_a^c + V_{st} \bar{\Sigma}_a^{st}}{V_f N_f}.$$

At a *fixed spectrum*, Eq. (4.16) is

$$k_\infty(\tilde{e}) = \frac{b_{fe} + \tilde{e}(b_{fi} - b_{fe})}{a_{fe} + \tilde{e}(a_{fi} - a_{fe}) + C}.$$

Solving for an assigned lattice target k_* gives

$$\tilde{e}_* = \frac{k_*(a_{fe} + C) - b_{fe}}{(b_{fi} - b_{fe}) - k_*(a_{fi} - a_{fe})}.$$

Match the enrichment basis: $\tilde{e}$ is an atom fraction; OpenMC's uranium `add_element` uses U-235 *weight percent*. Convert, or set isotope atom fractions explicitly.

OpenMC | Exploring fast spectra

Compare a fuel-cladding-Na hexagonal cell with its inventory-matched homogeneous mixture. Set `HexagonalPrism`'s `edge_length` to $P/\sqrt{3}$. Suppress axial and lateral leakage; compare regional spectra, regional reaction rates, and k_∞ .

Sources: Ch. 4, § 4.3; slide sets 16–17. OpenMC: `HexagonalPrism`; `Material.add_element`.